

HandsOn Knowledge Graphs and Graph AI for Biomedical Research

Learning goals

By the end of this workshop, I can:

- Explain why connected biomedical data suits graphs
- Run Neo4j with Docker
- Build a Drug–Gene–Disease graph
- Write and adapt Cypher queries
- Apply basic graph analytics
- Interpret a graph-based candidate link responsibly
- Ask an AI agent a research-grade graph question

Lab 0 — Environment check

1. Start Docker Desktop
2. In the workshop folder, run: `docker compose up -d`
3. Open `http://localhost:7474`
4. Login to Neo4j
5. Run: RETURN "Ready for Clinical Knowledge Graph Lab" AS status;

[Checkpoint: Screenshot your successful result.]

Lab 1 — My graph schema

Draw or annotate this:

Drug → TARGETS → Protein → ENCODED_BY → Gene Gene → ASSOCIATED_WITH → Disease Drug → TREATS → Disease

Questions:

1. Which relationships require an evidence score?
2. Which relationships require a source?
3. What additional entity would improve this graph? Pathway / Publication / Side effect / Clinical trial

Lab 2 — Import and validate

Run:

- `00_setup.cypher`
- `01_import_graph.cypher`
- `02_validate_graph.cypher`

Record:

- Number of Drug nodes:
- Number of Disease nodes:
- Number of Gene nodes:
- Number of TARGETS relationships:
- Does every relationship have a source?

Lab 3 — Cypher investigation

Choose one disease: []

Write your research question: []

Paste or write your Cypher query: []

What evidence path did you find? [Drug → Protein → Gene → Disease]

Evidence source(s): []

Lab 4 — Graph AI

Before running PageRank, predict the most influential node type: [Drug / Protein / Gene / Disease]

Result: []

One interpretation: []

One limitation: []

Lab 5 — Link prediction

Candidate drug: []

Candidate disease: []

Supporting genes: []

Prediction / network score: []

Why is this only a hypothesis and not a treatment claim? []

Lab 6 — AI research agent

Question asked: []

Cypher generated: []

Did the agent show sources and limitations? [Yes / No]

How would you improve the question? []

Team challenge

Present:

1. Disease investigated
2. One candidate research hypothesis
3. Evidence path
4. Source and score
5. Limitation
6. Next validation step